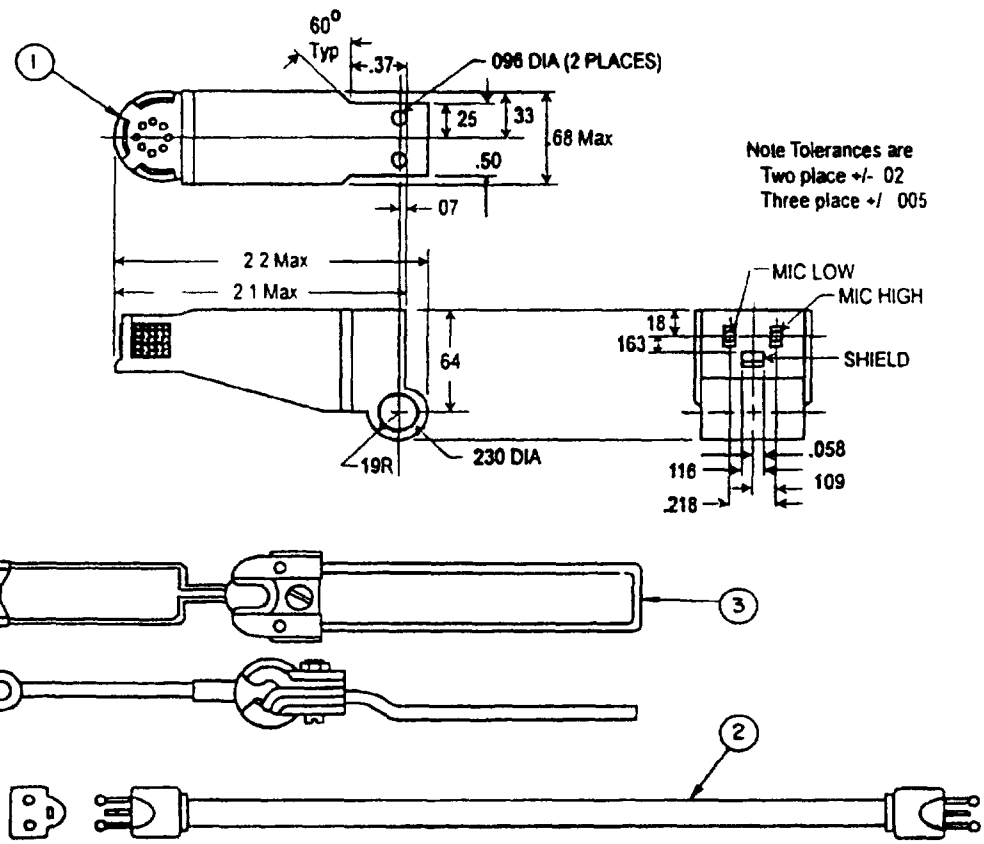
MILITARY SPECIFICATION SHEET

MICROPHONES, LINEAR M49199/3

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Department and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation:
MIL-M-49199



Find No.	Qty Req'd	Part No. or Identifying No	Nomenclature or Description
3	1	SM-B-883830	BOOM, MICROPHONE
2	1	DL D6 001 613	CORD ASSY, MICROPHONE (SEE PAGE 2)
1	1	M49199/3-*	MICROPHONE ELEMENT

FIGURE 1 Boom microphone assembly, P/N M49199/3-*

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

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REQUIREMENTS.

DESIGN and CONSTRUCTION See Figure 1

MICROPHONE ASSEMBLY (Microphone element, boom assembly and cable)

M49199/3-1. Cable assembly length is 13 inches
M49199/3-2 Cable assembly length is 16 inches

WEIGHT 62g max for the assembly

COLOR Semigloss, lusterless, black

JACKET MATERIAL Cable jacket material shall be vulcanized rubber

SOLDERABILITY Soldering shall be in accordance with MIL-STD-2000

VOLTAGE OPERATING RANGE 12 - 30 volts DC, 27 5 VDC nominal

INPUT CURRENT 17 mA max at 27 5 VDC when connected as shown in Figure 2

IMPEDANCE The electrical impedance of the microphone assembly shall be such that the unit shall be capable of meeting the sensitivity and frequency response to a close sound source while connected to a 100 ohm ±1% load impedance

SENSITIVITY 96 4 mV minmum at 104 dB SPL with 27 5 VDC supply voltage, and shall not vary by more than 1 0 dB when the supply voltage is varied between 22 to 30 volts

SENSITIVITY AT REDUCED BIAS VOLTAGE Shall not vary by more than 12dB at 104 dB SPL from the 27 5 VDC established value when the supply voltage is varied between 12 and 22 volts

SENSITIVITY AT ALTITUDE ±3dB of sensitivity at ground level while at a simulated altitude of 15,000 ft above sea level

FREQUENCY RESPONSE AT GROUND LEVEL See Figure 3

FREQUENCY RESPONSE AT ALTITUDE Within ±5dB of ground level measurement

NOISE IMMUNITY See Figure 4

LINEARITY Resistive load shall be 100 ±1% ohms

GAIN STABILIZATION

Within 100 mS when supply voltage is between 23 and 30 VDC

Within 150 mS when supply voltage is between 12 and 22 VDC

ENVIROMENTAL STRESS SCREENING (ESS) 24 hours of temperature cycling between -54°C and 55°C, with each cycle consisting of 1 hour at each temperature and the change between temperature occuring at a minimum rate of 5°C per minute

OVERVOLTAGE PROTECTION 15 minutes operation in test circuit shown in Figure 2 while the 330 ohm resistor is shorted

THERMAL SHOCK N/A

EXPLOSIVE ATMOSPHERE N/A

MARKING.

M49199/3-*
MANUFACTURER'S NAME OR CAGE CODE
CONTRACT NUMBER
DATE CODE
SERIAL NUMBER

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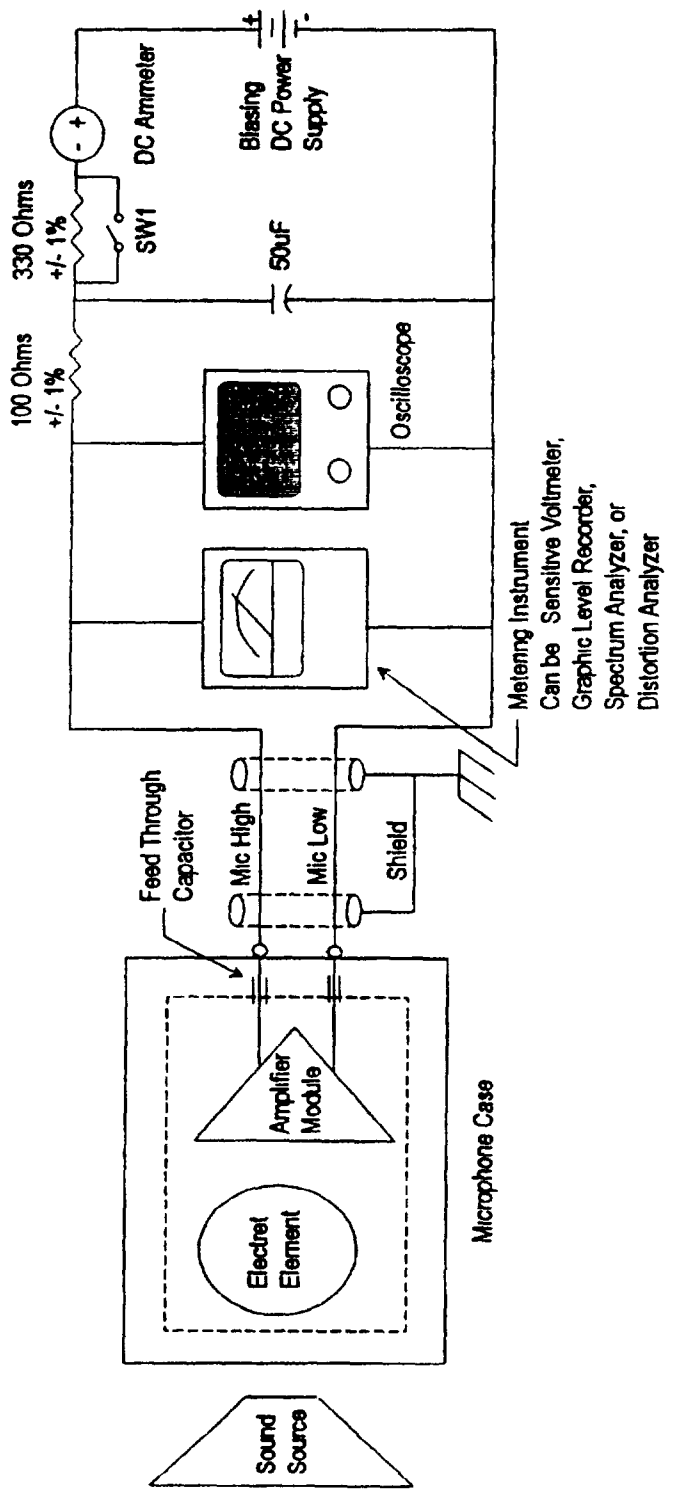


FIGURE 2. Test circuit schematic

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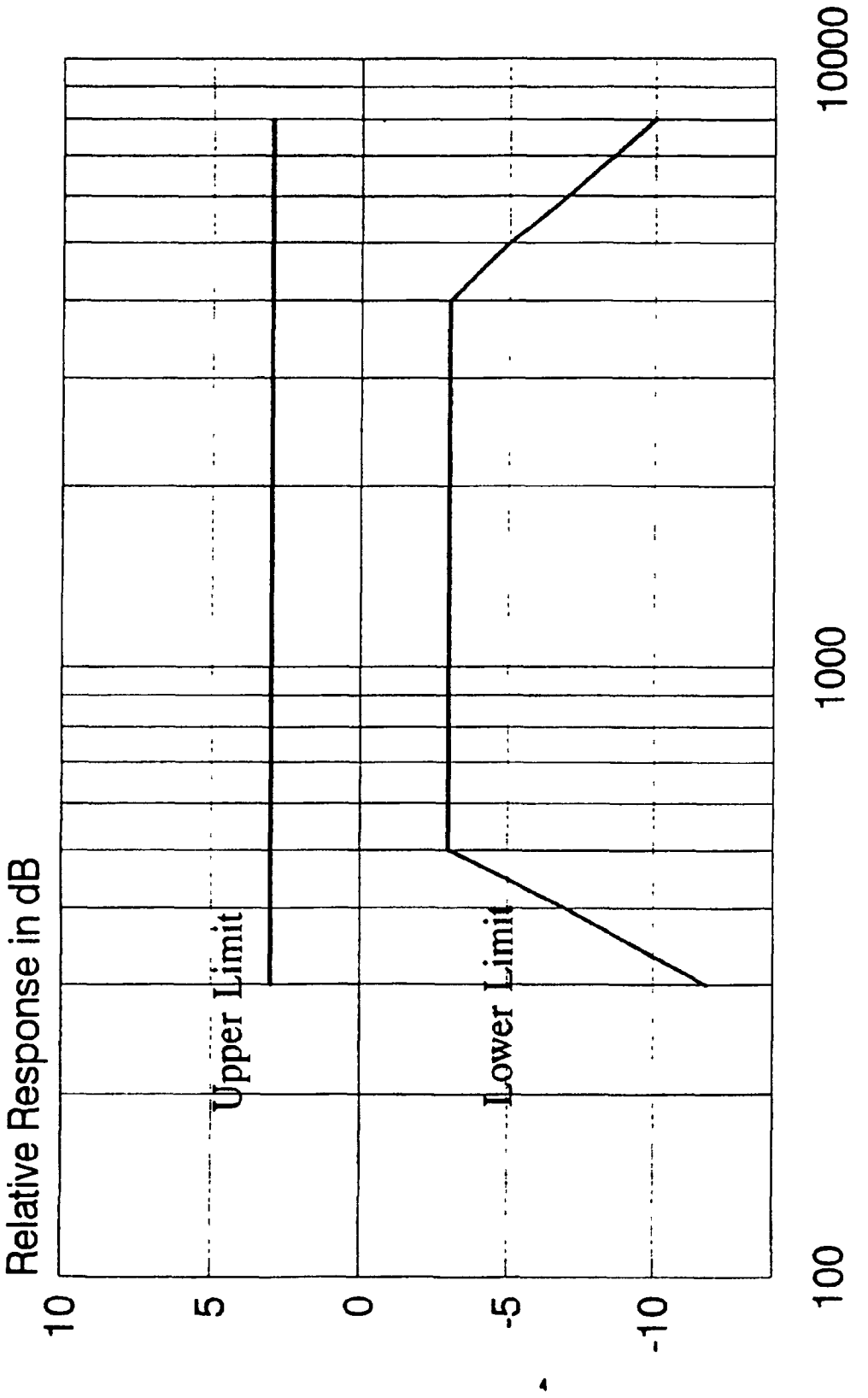


FIGURE 3 Frequency response limit curves

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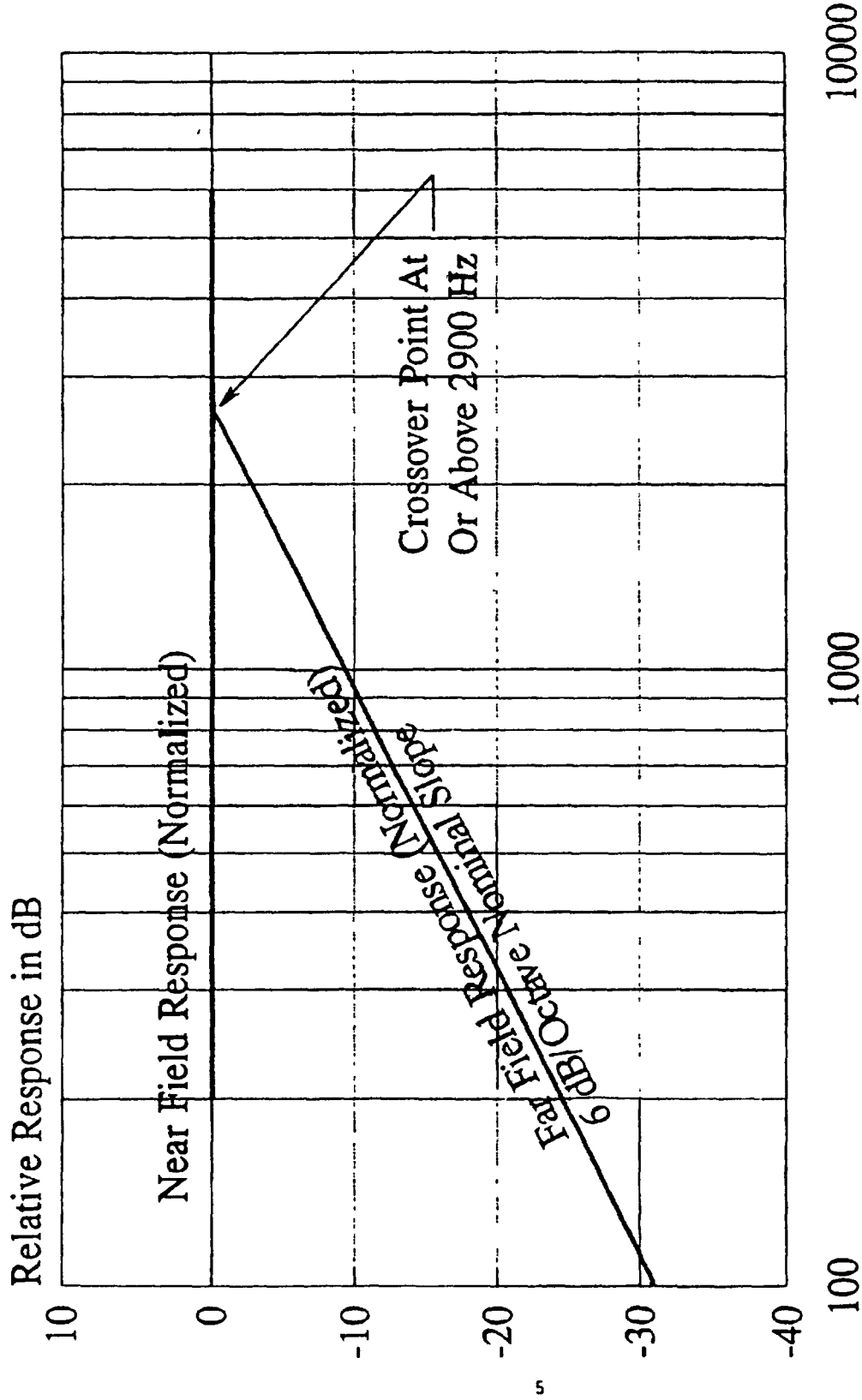


FIGURE 4 Noise-immunity characteristics

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TABLE I MIL-STD-461 RS03 electromagnetic environmental test levels

FREQUENCY	SIGNAL	MODULATION	FIELD 3/
14 KHz - 1 MHz	CW	NONE	20 V/M avg
1 MHz - 18 GHz	CW	NONE	200 V/M avg
18 GHz - 40 GHz	CW	NONE	20 V/M avg
1/ 1 MHz - 18 GHz	PAM	Table 1	200 V/M peak
18 GHz - 40 GHz	PAM	Table 1	20 V/M peak

TABLE II. Worst case modulation over required frequency band

STEP	FREQUENCY SUB-BAND	INITIAL SET FREQUENCY	FIELD (PEAK)	MODULATION (SQUARE WAVE)
A	1 0 MHz - 10 0 MHz	5 0 MHz	200 V/M	30 Hz, 300 Hz, 3 KHz, 30 KHz
B	10 0 MHz - 100 0 MHz	50 0 MHz	200 V/M	30 Hz, 300 Hz, 3 KHz, 30 KHz
C	100 0 MHz - 1 0 GHz	500 0 MHz	200 V/M	30 Hz, 300 Hz, 3 KHz, 30 KHz
D	1 0 GHz - 10 0 GHz	SEE D_1 AND D_2 WHERE APPLICABLE		
D_1	1 0 GHz - 8 0 GHz	5 0 GHz	200 V/M	3 KHz 30 KHz, 300 KHz
D_2	8 0 GHz - 10 0 GHz	9 0 GHz	200 V/M	3 KHz, 30 KHz 300 KHz
E	10 0 GHz - 18 0 GHz	15 0 GHz	200 V/M	3 KHz 30 KHz 300 KHz
F	18 0 GHz - 40 GHz	30 0 GHz	20 V/M	300 Hz, 3 KHz, 30 KHz

TABLE III Modulation fall back values

SQUARE WAVE (50% DUTY CYCLE)
300 Hz
300 Hz
250 Hz
See D_1 or D_2
100 KHz
250 KHz
60 KHz
2 KHz

NOTES:

1/ Tests from 1 0 MHz - 18 0 GHz and 18 0 GHz - 40 0 GHz shall be performed by sub-bands defined in Table II for worst case modulation conditions If no susceptibility exists in the sub-band specified, the modulation in Table III shall be used Note 2/ below, defines the procedures for determining modulation required

2/ EXAMPLE Step A (1 0 - 10 0 MHz)
(a) Adjust to initial set frequency of 5 0 MHz
(b) Perform RS03 Tests @ 200 V/M for each modulation specified in Table II
(c) Using worst case modulation determined in STEP (b), perform tests over frequency sub-bands @ 200 V/M If no susceptibility condition was encountered in STEP (b), use modulation specified in TABLE III for 1 0 - 10 0 MHz

3/ NOTE

$$\frac{V}{M}(P)\sqrt{DutyCycle} = \frac{V}{M}(AV)$$

For CW

$$V/M\ (P)\,rms = V/M\ (AV)\,rms$$

4/ Except for 33-34 GHz where the field peak intensity is 200 V/M

MIL-M-49199/3(AS)

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INTENDED USE. Microphone, identified as linear, M49199/3-*, is a noise-canceling, DC-powered, high-gain microphone designed for boom mount use on Navy fixed- and rotary- wing helmets The microphone is intended for use as a part of the airborne communication system which will provide communication of high intelligibility under the extreme noise conditions encountered in military aircraft The microphone is designed to work with intercommunication system AIC-14, specifically control box C-2645D The preservation, packing, and marking specified herein are intended for direct shipments to the Government However, this specification may also be used for the preparation of microphones for shipment from the parts contractor to the original equipment manufacturer

Preparing Activity
Navy - AS
(Project 5965-0220-3)

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